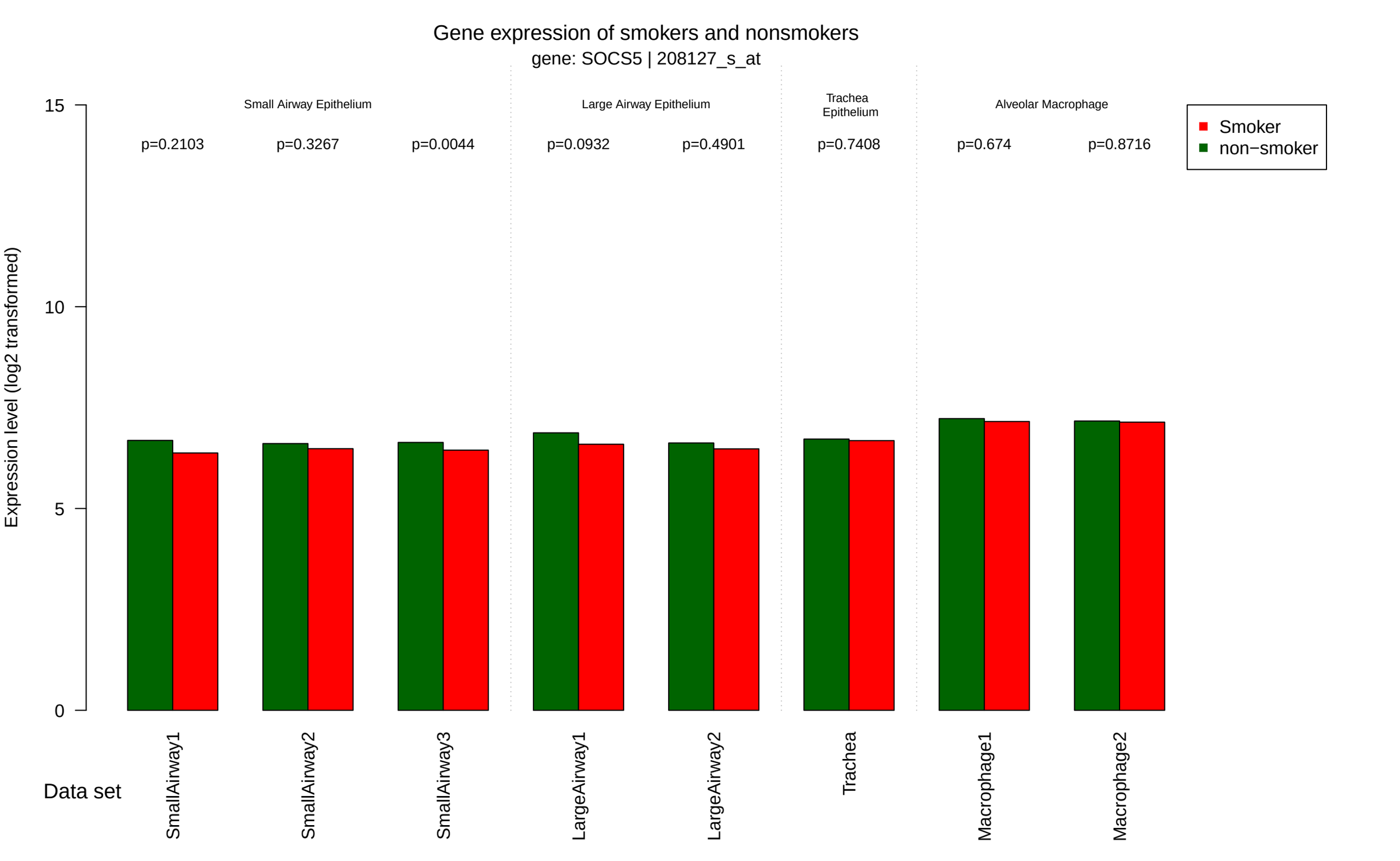
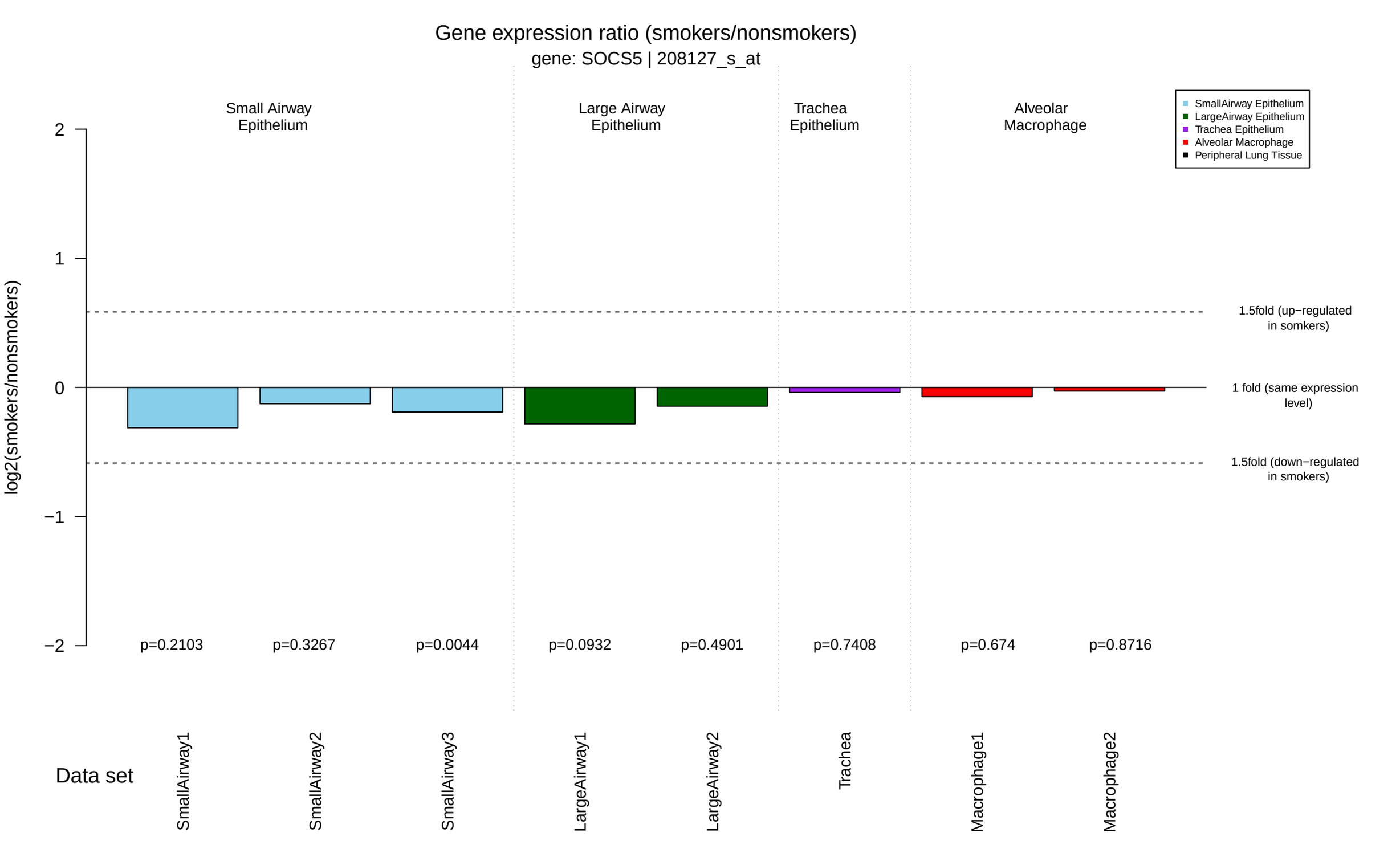
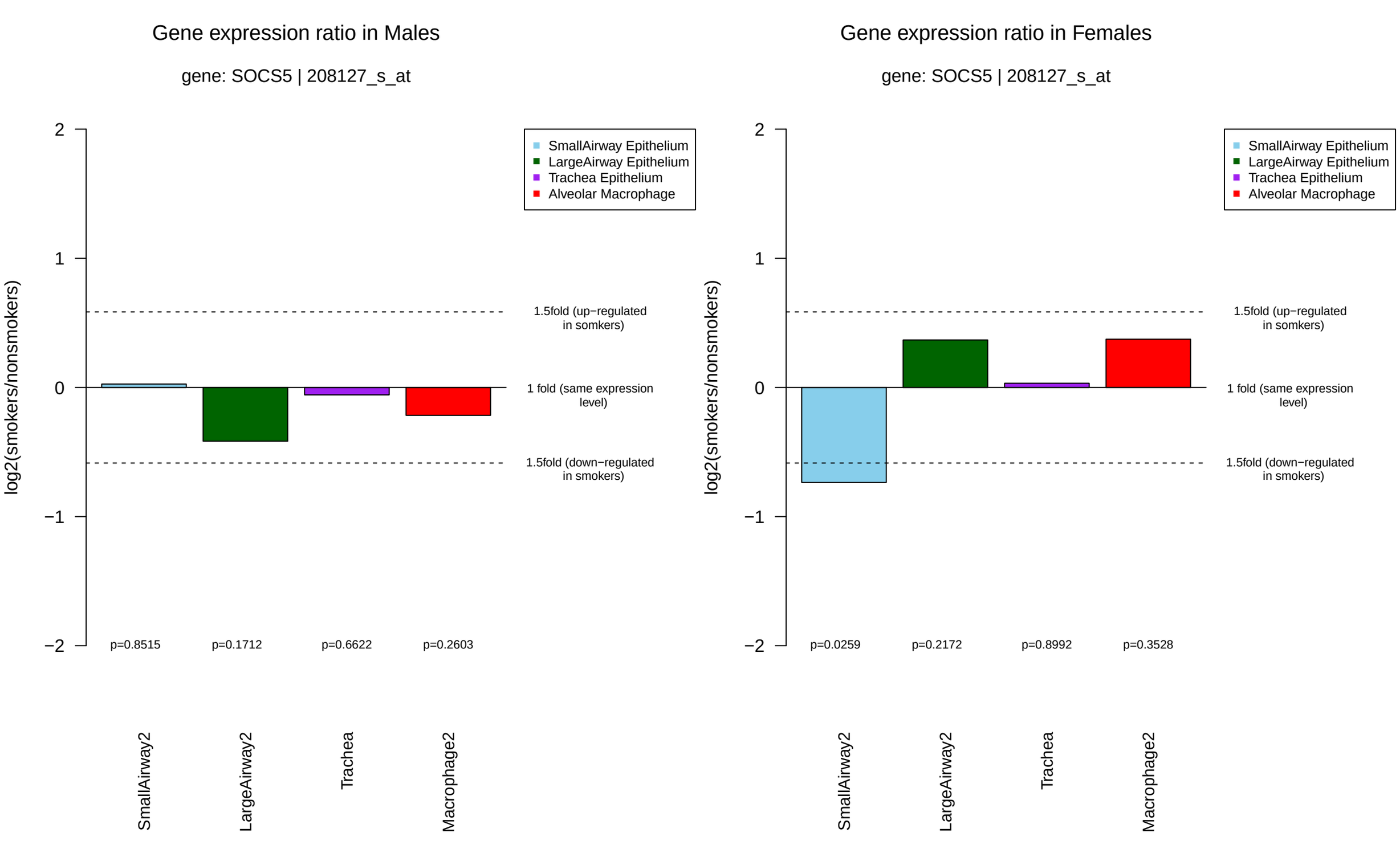
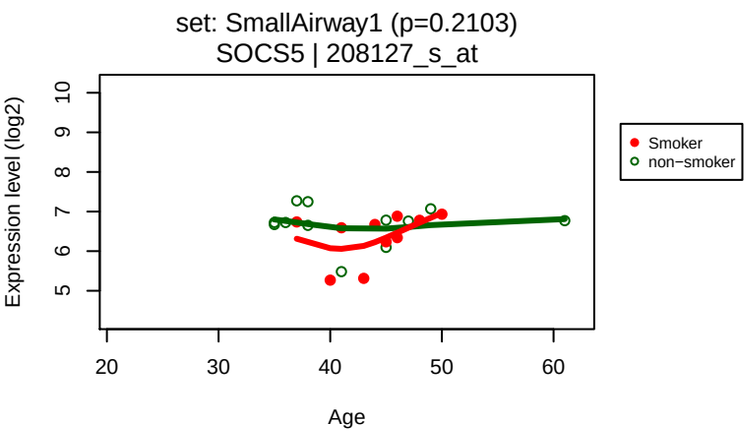
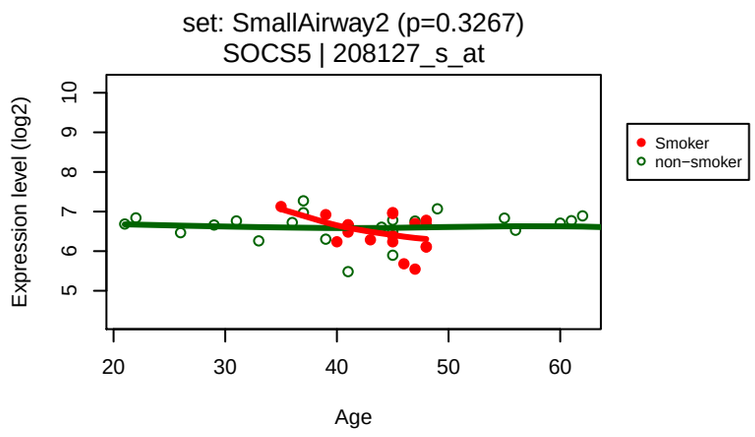
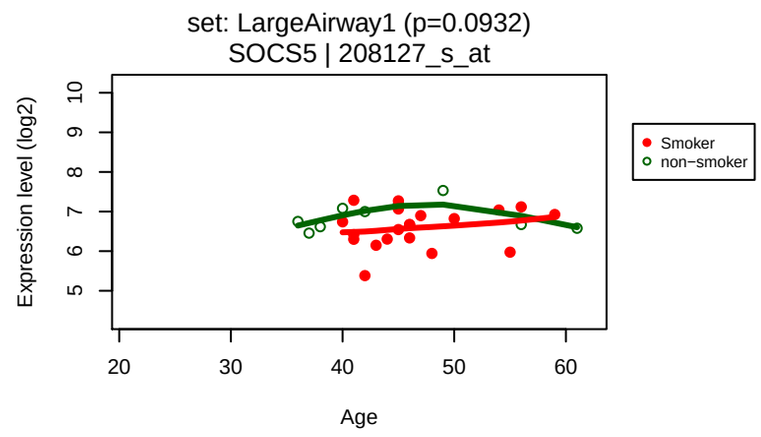

Gene expression of smokers and nonsmokers
gene: SOCS5 | 208127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2103
p=0.3267
p=0.0044
p=0.0932
p=0.4901
p=0.7408
p=0.674
p=0.8716
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

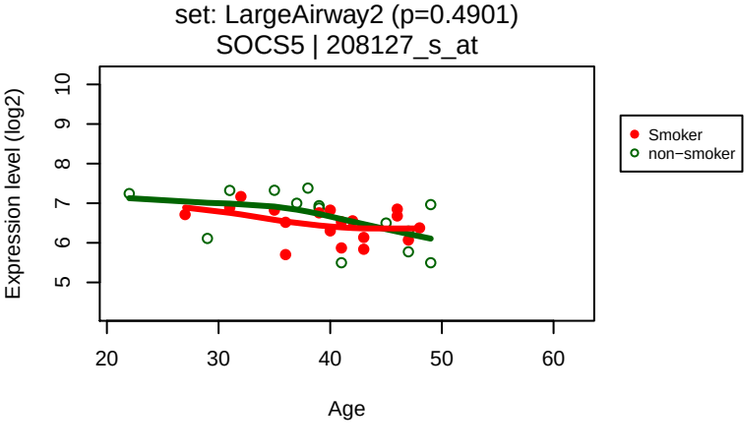

Gene expression ratio (smokers/nonsmokers)
gene: SOCS5 | 208127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2103
p=0.3267
p=0.0044
p=0.0932
p=0.4901
p=0.7408
p=0.674
p=0.8716
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

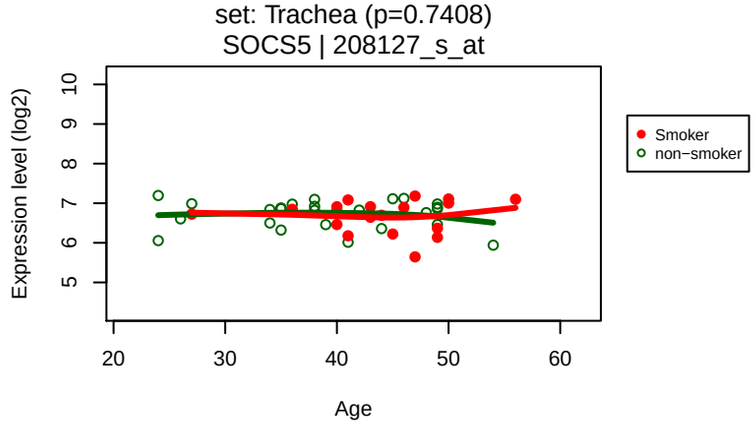

Gene expression ratio in Males
gene: SOCS5 | 208127_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8515
p=0.1712
p=0.6622
p=0.2603
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS5 | 208127_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0259
p=0.2172
p=0.8992
p=0.3528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

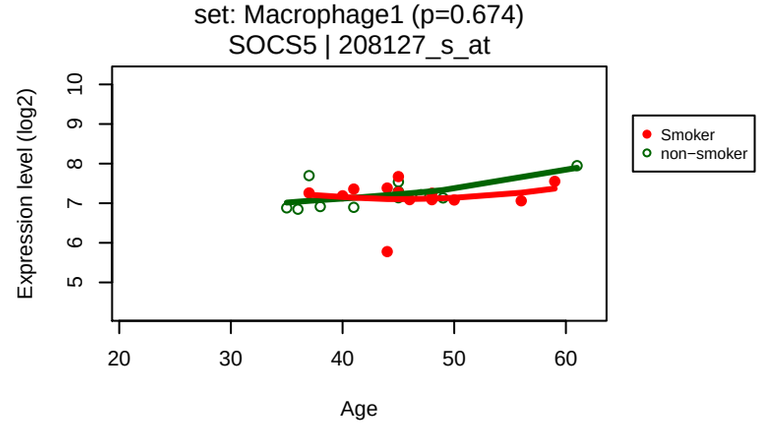

set: SmallAirway1 (p=0.2103)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker

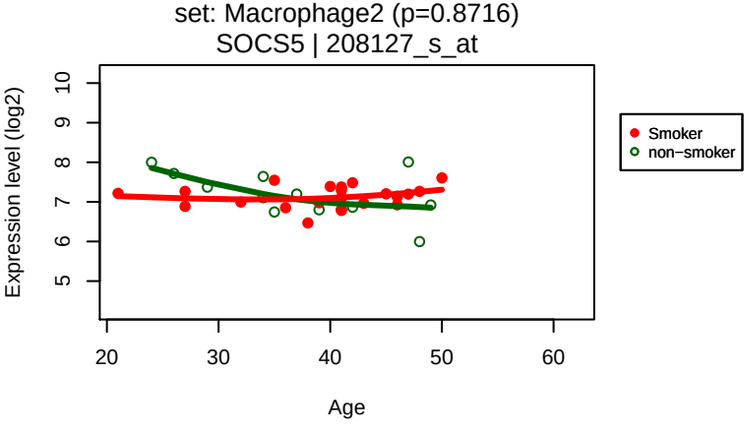

set: SmallAirway2 (p=0.3267)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker

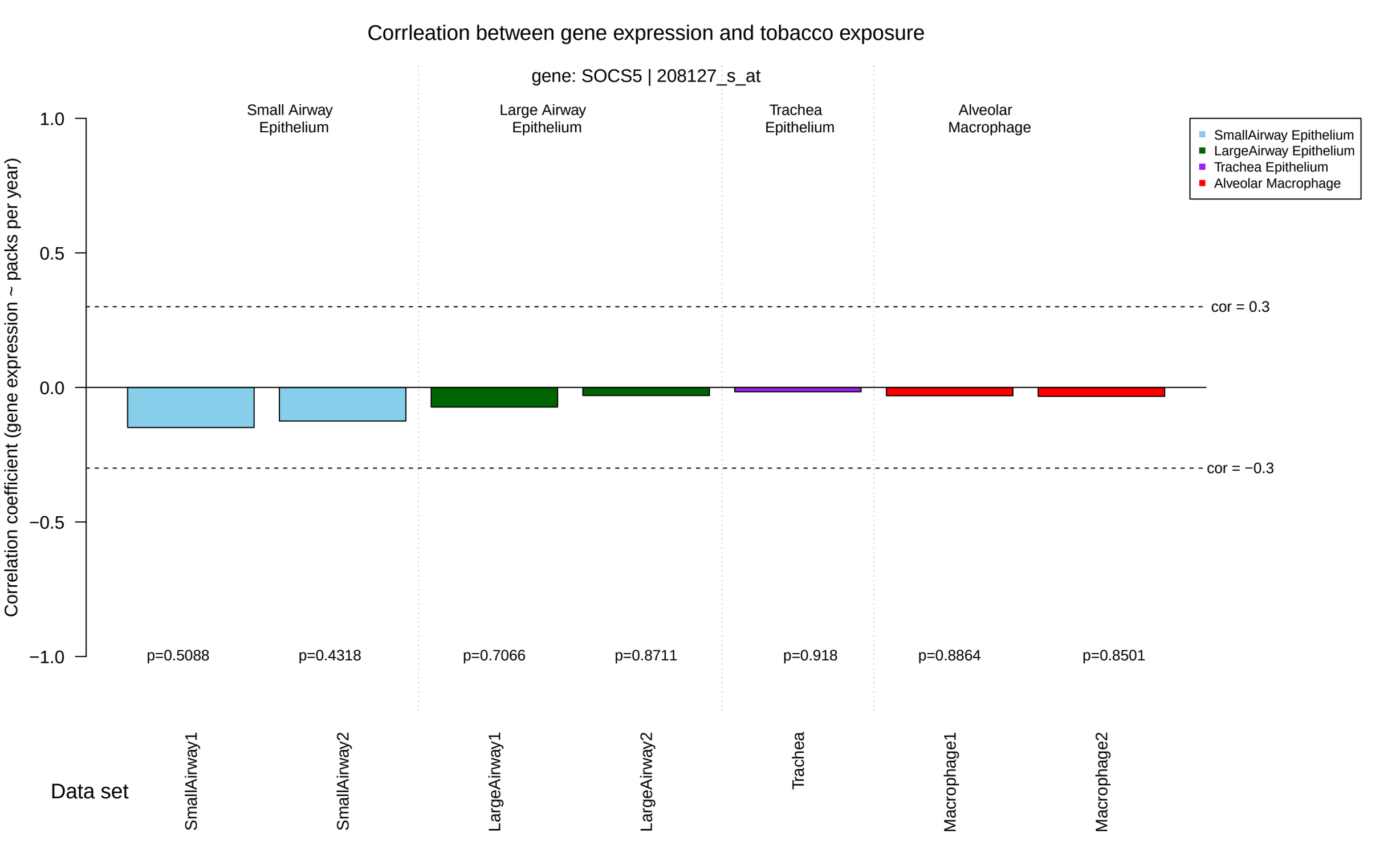

set: LargeAirway1 (p=0.0932)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4901)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7408)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.674)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8716)
SOCS5 | 208127_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS5 | 208127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5088
p=0.4318
p=0.7066
p=0.8711
p=0.918
p=0.8864
p=0.8501
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2